

Лабораторная работа № 4. Первоначальное конфигурирование сети

Абакумова О. М.

Российский университет дружбы народов, Москва, Россия

Информация

- Абакумова Олеся Максимовна
- Студентка
- Российский университет дружбы народов
- 1132220832@pfur.ru
- <https://github.com/omabakumova>



Цель работы

Провести подготовительную работу по первоначальной настройке коммутаторов сети.

Задания

Требуется сделать первоначальную настройку коммутаторов сети. Под первоначальной настройкой понимается указание имени устройства, его IP-адреса, настройка доступа по паролю к виртуальным терминалам и консоли, настройка удалённого доступа к устройству по ssh.

Выполнение лабораторной работы

Первоначальная настройка коммутаторов сети

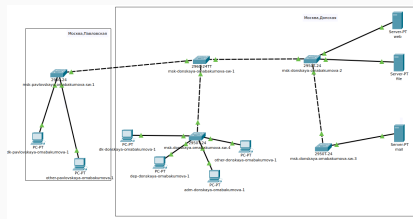


Рис. 1: Размещение коммутаторов и конечных устройств согласно схеме сети L1

Первоначальная настройка коммутаторов сети

```
Switch(config)#hostname msk-donskaya-sw-1
msk-donskaya-sw-1(config)#interface vlan2
msk-donskaya-sw-1(config-if)#no shutdown
msk-donskaya-sw-1(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-sw-1(config-if)#exit
msk-donskaya-sw-1(config)#ip default-gateway 10.128.1.1
^
% Invalid input detected at '^' marker.

msk-donskaya-sw-1(config)#ip default-gateway 10.128.1.1
msk-donskaya-sw-1(config)#line vty 0 4
msk-donskaya-sw-1(config-line)#password cisco
msk-donskaya-sw-1(config-line)#login
msk-donskaya-sw-1(config-line)#exit
msk-donskaya-sw-1(config)#line console 0
msk-donskaya-sw-1(config-line)#password cisco
msk-donskaya-sw-1(config-line)#login
msk-donskaya-sw-1(config-line)#exit
msk-donskaya-sw-1(config)#enable secret cisco
msk-donskaya-sw-1(config)#service password-encryption
msk-donskaya-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-sw-1(config)#line vty 0 4
*Mar 1 0:17:26.698: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:17:26.698: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-sw-1(config-line)#transport input ssh
msk-donskaya-sw-1(config-line)#exit
msk-donskaya-sw-1(config)#exit
msk-donskaya-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-sw-1#
```

Рис. 2: Настройка msk-donskaya-omabakumova-sw-1

Первоначальная настройка коммутаторов сети

```
Switch(config)#hostname msk-donskaya-omabakumova-sw-2
msk-donskaya-omabakumova-sw-2(config)#interface vlan2
msk-donskaya-omabakumova-sw-2(config-if)#no shutdown
msk-donskaya-omabakumova-sw-2(config-if)#ip address 10.128.1.1 255.255.255.0
msk-donskaya-omabakumova-sw-2(config-if)#exit
msk-donskaya-omabakumova-sw-2(config)#ip default-gateway 10.128.1.1
msk-donskaya-omabakumova-sw-2(config)#line vty 0 4
msk-donskaya-omabakumova-sw-2(config-line)#password cisco
msk-donskaya-omabakumova-sw-2(config-line)#login
msk-donskaya-omabakumova-sw-2(config-line)#exit
msk-donskaya-omabakumova-sw-2(config)#line console 0
msk-donskaya-omabakumova-sw-2(config-line)#password cisco
msk-donskaya-omabakumova-sw-2(config-line)#login
msk-donskaya-omabakumova-sw-2(config-line)#exit
msk-donskaya-omabakumova-sw-2(config)#enable secret cisco
msk-donskaya-omabakumova-sw-2(config)#service password-encryption
msk-donskaya-omabakumova-sw-2(config)#username admin privilege 1 secret cisco
msk-donskaya-omabakumova-sw-2(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-omabakumova-sw-2(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-omabakumova-sw-2.donskeya.rudn.edu
Choose the size of the key modulus in the range of 300 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-omabakumova-sw-2(config)#line vty 0 4
*Mar 1 0:24:30.330: %SSH-5-ENABLED: SSH 1.99 has been enabled
msk-donskaya-omabakumova-sw-2(config-line)#transport input ssh
msk-donskaya-omabakumova-sw-2(config-line)#exit
msk-donskaya-omabakumova-sw-2#
%SYS-5-CONFIG-I: Configured from console by console

msk-donskaya-omabakumova-sw-2#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-sw-2#
```

Рис. 3: Настройка msk-donskaya-omabakumova-sw-2

Первоначальная настройка коммутаторов сети

```
Switch(config)#hostname msk-donskaya-omabakumova-sw-3
msk-donskaya-omabakumova-sw-3(config)#interface vlan2
msk-donskaya-omabakumova-sw-3(config-if)#no shutdown
msk-donskaya-omabakumova-sw-3(config-if)#ip address 10.128.1.4 255.255.255.0
msk-donskaya-omabakumova-sw-3(config-if)#exit
msk-donskaya-omabakumova-sw-3(config)#ip default-gateway 10.128.1.1
msk-donskaya-omabakumova-sw-3(config)#line vty 0 4
msk-donskaya-omabakumova-sw-3(config-line)#password cisco
msk-donskaya-omabakumova-sw-3(config-line)#login
msk-donskaya-omabakumova-sw-3(config-line)#exit
msk-donskaya-omabakumova-sw-3(config)#line console 0
msk-donskaya-omabakumova-sw-3(config-line)#password cisco
msk-donskaya-omabakumova-sw-3(config-line)#login
msk-donskaya-omabakumova-sw-3(config-line)#exit
msk-donskaya-omabakumova-sw-3(config)#enable secret cisco
msk-donskaya-omabakumova-sw-3(config)#service password-encryption
msk-donskaya-omabakumova-sw-3(config)#username admin privilege 1 secret
% Incomplete command.
msk-donskaya-omabakumova-sw-3(config)#username admin privilege 1 secret cisco
msk-donskaya-omabakumova-sw-3(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-omabakumova-sw-3(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-omabakumova-sw-3.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-omabakumova-sw-3(config)#line vty 0 4
*Mar 1 0:20:14.485: %SSH-5-ENABLED: SSH 1.90 has been enabled
msk-donskaya-omabakumova-sw-3(config-line)#transport input ssh
msk-donskaya-omabakumova-sw-3(config-line)#exit
msk-donskaya-omabakumova-sw-3(config)#exit
msk-donskaya-omabakumova-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-3#write memory
Building Configuration...
[OK]
msk-donskaya-omabakumova-sw-3#
```

Рис. 4: Настройка msk-donskaya-omabakumova-sw-3

Первоначальная настройка коммутаторов сети

```
Switch(config)#hostname msk-donskaya-omabakumova-sw-4
msk-donskaya-omabakumova-sw-4(config)#interface vlan2
msk-donskaya-omabakumova-sw-4(config-if)#no shutdown
msk-donskaya-omabakumova-sw-4(config-if)#ip address 10.128.1.5 255.255.255.0
msk-donskaya-omabakumova-sw-4(config-if)#exit
msk-donskaya-omabakumova-sw-4(config)#ip default-gateway 10.128.1.1
msk-donskaya-omabakumova-sw-4(config)#line vty 0 4
msk-donskaya-omabakumova-sw-4(config-line)#password cisco
msk-donskaya-omabakumova-sw-4(config-line)#login
msk-donskaya-omabakumova-sw-4(config-line)#exit
msk-donskaya-omabakumova-sw-4(config)#line console 0
msk-donskaya-omabakumova-sw-4(config-line)#password cisco
msk-donskaya-omabakumova-sw-4(config-line)#login
msk-donskaya-omabakumova-sw-4(config-line)#exit
msk-donskaya-omabakumova-sw-4(config)#enable secret cisco
msk-donskaya-omabakumova-sw-4(config)#service password-encryption
msk-donskaya-omabakumova-sw-4(config)#username admin privilege 1 secret cisco
msk-donskaya-omabakumova-sw-4(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-omabakumova-sw-4(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-omabakumova-sw-4.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-omabakumova-sw-4(config)#line vty 0 4
*Mar 1 0:35:54.722: %SSH-5-ENABLED: SSH 1.99 has been enabled
msk-donskaya-omabakumova-sw-4(config-line)#transport input ssh
msk-donskaya-omabakumova-sw-4(config-line)#exit
msk-donskaya-omabakumova-sw-4(config)#exit
msk-donskaya-omabakumova-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-4#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-sw-4#
```

Рис. 5: Настройка msk-donskaya-omabakumova-sw-4

Первоначальная настройка коммутаторов сети

```
Switch(config)#hostname msk-pavlovskaya-omabakumova-sw-1
msk-pavlovskaya-omabakumova-sw-1(config)#interface vlan2
msk-pavlovskaya-omabakumova-sw-1(config-if)#no shutdown
msk-pavlovskaya-omabakumova-sw-1(config-if)#ip address 10.128.1.6 255.255.255.0
msk-pavlovskaya-omabakumova-sw-1(config-if)#exit
msk-pavlovskaya-omabakumova-sw-1(config)#ip default-gateway 10.128.1.1
msk-pavlovskaya-omabakumova-sw-1(config)#line vty 0 4
msk-pavlovskaya-omabakumova-sw-1(config-line)#password cisco
msk-pavlovskaya-omabakumova-sw-1(config-line)#login
msk-pavlovskaya-omabakumova-sw-1(config-line)#exit
msk-pavlovskaya-omabakumova-sw-1(config)#enable secret cisco
msk-pavlovskaya-omabakumova-sw-1(config)#line console 0
^
% Invalid input detected at '^' marker.

msk-pavlovskaya-omabakumova-sw-1(config)#line console 0
^
% Invalid input detected at '^' marker.

msk-pavlovskaya-omabakumova-sw-1(config)#line console 0
msk-pavlovskaya-omabakumova-sw-1(config-line)#password cisco
msk-pavlovskaya-omabakumova-sw-1(config-line)#login
msk-pavlovskaya-omabakumova-sw-1(config-line)#exit
msk-pavlovskaya-omabakumova-sw-1(config)#enable secret cisco
msk-pavlovskaya-omabakumova-sw-1(config)#service password-encryption
msk-pavlovskaya-omabakumova-sw-1(config)#username admin privilege 1 secret cisco
msk-pavlovskaya-omabakumova-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-pavlovskaya-omabakumova-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-pavlovskaya-omabakumova-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-omabakumova-sw-1(config)#line vty 0 4
*Mar 1 0:47:28.0: %SSH-5-ENABLED: SSH 1.00 has been enabled
msk-pavlovskaya-omabakumova-sw-1(config-line)#transport input ssh
msk-pavlovskaya-omabakumova-sw-1(config-line)#exit
msk-pavlovskaya-omabakumova-sw-1(config)#exit
msk-pavlovskaya-omabakumova-sw-1#
%SYS-5-CONFIG-I: Configured from console by console

msk-pavlovskaya-omabakumova-sw-1#write memory
Building configuration...
[OK]
msk-pavlovskaya-omabakumova-sw-1#
```

Рис. 6: Настройка msk-pavlovskaya-omabakumova-sw-1

Первоначальная настройка коммутаторов сети

```
User Access Verification
Password:
msk-donskaya-sw-1>enable
Password:
msk-donskaya-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-sw-1(config)#hostname msk-donskaya-onabakumova-sw-1
msk-donskaya-onabakumova-sw-1(config)#exit
msk-donskaya-onabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-onabakumova-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-onabakumova-sw-1#
```

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Рис. 7: Выполнение соглашения об именовании

Выводы

Во время выполнения данной лабораторной работы я провела подготовительную работу по первоначальной настройке коммутаторов сети.